

COMPUTER ORGANIZATION PRACTICE SHEET

NIMCET TARGET PAPER • COMPUTER BASICS

ResourceOS

1. Which of the following is true regarding Von Neumann architecture?

- (A) Separate storage for inputs and outputs
- (B) Separate memory for data and instructions
- (C) Single memory for both data and instructions
- (D) Separate processing units for instructions

2. The register that holds the address of the memory location where the next instruction is located is:

- (A) Memory Data Register
- (B) Program Counter
- (C) Instruction Register
- (D) Memory Address Register

3. How many RAM chips of capacity $4K \times 8$ -bit are needed to provide a memory capacity of 64 Kbytes?

- (A) 8
- (B) 16
- (C) 32
- (D) 64

4. How many separate address and data lines are needed for a memory unit of $4K \times 16$ capacity?

- (A) 10 address, 16 data
- (B) 12 address, 16 data
- (C) 12 address, 8 data
- (D) 10 address, 12 data

5. Which of the following components is NOT a part of an instruction format in CPU processing?

- (A) Opcode
- (B) Source operand
- (C) Register file
- (D) Destination operand

6. The interface that transfers data between memory and I/O devices without involving the CPU is:

- (A) Serial Interface
- (B) Direct Memory Access (DMA)
- (C) Instruction Register
- (D) Programmed I/O

7. What is the primary purpose of cache memory in a computer system?

- (A) Manage I/O operations
- (B) Permanently store data and programs
- (C) Temporarily store frequently accessed data
- (D) Provide secondary disk space

8. The time required for fetching and execution of one machine instruction is called:

- (A) Seek time
- (B) CPU cycle
- (C) Real time
- (D) Delay time

9. Cache memory functions as an intermediary directly between:

- (A) RAM and ROM
- (B) CPU and RAM
- (C) CPU and Hard Disk
- (D) Hard Disk and RAM

10. Which component establishes a communication link between the CPU and peripheral devices?

- (A) Memory Address Register
- (B) Interface Unit / Controller
- (C) Instruction Register
- (D) Program Counter

11. A CPU generates 32-bit virtual addresses with a page size of 4 KB. How many bits are used for the page offset?

- (A) 10 bits
- (B) 12 bits
- (C) 20 bits
- (D) 16 bits

12. Which register stores the intermediate results generated by the ALU?

- (A) Instruction Register
- (B) Accumulator
- (C) Program Counter
- (D) Memory Address Register

13. Cache memory relies on which of the following principles for its effective performance?

- (A) Locality of Reference
- (B) Direct Mapping
- (C) Static Allocation
- (D) Serial Processing

14. How many 32K × 1 RAM chips are needed to construct a total memory capacity of 256 Kbytes?

- (A) 8
- (B) 32
- (C) 64
- (D) 128

15. Which signal bus in a standard CPU architecture is unidirectional?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) System Bus

16. The main memory type that requires periodic electrical refreshing of data stored in capacitors is:

- (A) SRAM
- (B) DRAM
- (C) ROM
- (D) EEPROM

17. In a processor, what is the exact order of execution in an instruction pipeline cycle?

- (A) Execute → Fetch → Decode
- (B) Fetch → Decode → Execute
- (C) Decode → Fetch → Execute
- (D) Fetch → Execute → Decode

18. Which storage buffer compensates for differences in data flow rates between two hardware devices?

- (A) Cache
- (B) Buffer
- (C) Accumulator
- (D) Register

19. Upon receiving an interrupt signal from an external I/O device, the CPU:

- (A) Restarts the system immediately
- (B) Branches to Interrupt Service Routine after current instruction execution
- (C) Ignores the signal until program completion
- (D) Clears all general-purpose registers

20. Which register keeps track of the top memory address in a stack structure?

- (A) Memory Data Register
- (B) Stack Pointer
- (C) Index Register
- (D) Instruction Register

21. A system uses 16-bit wide memory addressing. What is the maximum number of addressable memory locations?

- (A) 32,768
- (B) 65,536
- (C) 1,048,576
- (D) 16,000

22. Computer architecture with separate physical memory units and buses for data and instructions is:

- (A) Von Neumann Architecture
- (B) Harvard Architecture
- (C) Princeton Architecture
- (D) Turing Architecture

23. Which register holds the actual data or word being read from or written into the main memory?

- (A) Program Counter
- (B) Memory Data Register
- (C) Instruction Register
- (D) Index Register

24. Which memory unit provides the fastest access time for processor operations?

- (A) Main Memory (RAM)
- (B) Cache Memory
- (C) CPU Registers
- (D) Solid State Drive

25. In a microprogrammed control unit, control signals are generated by executing microinstructions stored in:

- (A) Main RAM
- (B) Control Memory
- (C) Hard Disk
- (D) Registers

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(C)	6	(B)	11	(B)	16	(B)	21	(B)
2	(B)	7	(C)	12	(B)	17	(B)	22	(B)
3	(B)	8	(B)	13	(A)	18	(B)	23	(B)
4	(B)	9	(B)	14	(C)	19	(B)	24	(C)
5	(C)	10	(B)	15	(B)	20	(B)	25	(B)